MATHEMATICS CORNER

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Misconception: Mathematics Is All About Formulas

THE MISCONCEPTION

"Mathematics is about formulas. Once you forget a formula, you will be unable to solve the question."

THE TRUTH

Mathematics is not merely about memorising formulas; it is about understanding concepts, recognising patterns, reasoning logically, and applying knowledge to solve problems. Formulas are useful tools, but they are not the whole of mathematics.

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Why is this a misconception?

Many learners believe that success in mathematics depends on remembering every formula their teacher introduces. Consequently, when they forget a formula during an examination, they panic and conclude that they cannot solve the question.

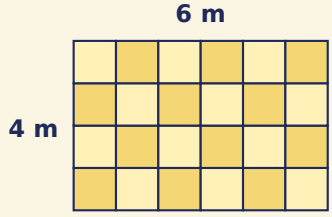
However, a learner who understands the concept behind a formula can often reconstruct the formula, use another method, or reason their way to the answer.

For example, consider the formula for the area of a rectangle:

$$A = l \times b$$

A learner may forget this formula but still understand what area means: the amount of surface covered by a shape.

Suppose a rectangular floor is 6 metres long and 4 metres wide. The learner can visualise six rows of four square metres, or four rows of six square metres.



Therefore,

$$A = 6 \times 4 = 24 \text{ m}^2$$

Even if the learner forgets the formula, an understanding of the concept of area can help them solve the problem.

Another illustration: The perimeter of a rectangle

Suppose a rectangle has a length of 8 cm and a width of 3 cm.

The formula is:

$$P = 2(l + b)$$

But what if a learner forgets it?

Instead of giving up, the learner can reason that perimeter is the total distance around a shape.

The four sides are 8 cm, 3 cm, 8 cm and 3 cm.

Therefore,

$$P = 8 + 3 + 8 + 3 = 22 \text{ cm}$$

The learner has solved the problem without recalling the formula.

This demonstrates that understanding gives learners alternative ways to solve problems, while memorisation alone can leave them helpless when recall fails.

Does this mean formulas are not important?

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Absolutely not! Formulas are important because they make calculations easier, faster and more efficient. The problem is not memorising formulas; the problem is memorising them without understanding what they mean, how they are derived, or when they should be used.

A good mathematics learner should aim to:

- Understand the concept behind a formula.
- Know what each symbol in the formula represents.
- Understand why the formula works.
- Learn how to derive or reconstruct simple formulas where possible.
- Practise applying formulas to different situations.
- Develop alternative problem-solving strategies.

What should mathematics teachers do?

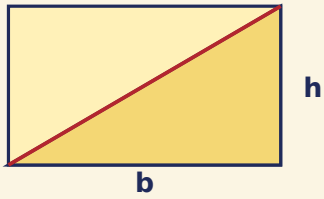
Teachers should not make mathematics lessons revolve solely around writing formulas on the board and asking learners to substitute values into them.

Instead, they should help learners discover mathematical relationships through diagrams, practical activities, discussions, patterns and real-life situations.

For instance, before introducing the formula for the area of a triangle,

$$A = \frac{1}{2} b h$$

a teacher could draw a rectangle, divide it diagonally into two equal triangles, and help learners see why each triangle has half the area of the rectangle.



When learners understand this relationship, the formula becomes meaningful rather than something they must memorise blindly.

Practical suggestions for JHS mathematics teachers

Ideas are only useful when they can be used on Monday morning. The twelve suggestions below are practical, need little or no money, and fit the JHS Common Core Programme. Each one comes with worked examples you can copy straight into your lessons.

1 Ask “Why does it work?” before “How do I do it?”

Before giving a rule, let learners discover it with paper, objects, measuring or drawing. A rule they have discovered is a rule they will remember, and one they can rebuild.

EXAMPLE 1 Angles in a triangle

Learners cut out any triangle, tear off its three corners and place them side by side on a straight line. The corners always make a straight line, so the angles add up to 180°.

Use it: two angles of a triangle are 50° and 60°. Find the third angle.

$$180^\circ - (50^\circ + 60^\circ) = 180^\circ - 110^\circ = 70^\circ$$

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EXAMPLE 2 The circumference of a circle

Learners measure the distance round a cup (circumference) and the distance across it (diameter), then divide. Every round object gives an answer close to 3.14, which is π.

Use it: a cup has a diameter of 7 cm. Find its circumference, using π = 22/7.

$$C = \pi d = \frac{22}{7} \times 7 = 22 \text{ cm}$$

EXAMPLE 3 The area of a parallelogram

Cut a triangle off one end of a paper parallelogram and slide it to the other end. You get a rectangle with the same base and height, so area = base × height.

Use it: base 9 cm and height 5 cm.

$$A = 9 \times 5 = 45 \text{ cm}^2$$

2 Teach learners to rebuild a forgotten formula

Set “rebuild” tasks regularly: “Pretend you have forgotten the formula. How can you still get the answer?” This is the habit that rescues learners in an examination.

EXAMPLE 4 Volume of a cuboid (5 cm by 4 cm by 3 cm)

Think of cubes. The bottom layer holds 5 × 4 = 20 cubes. There are 3 layers.

$20 \times 3 = 60 \text{ cm}^3$ (this is the formula $V = l \times b \times h$)

EXAMPLE 5 Area of a trapezium: parallel sides 6 cm and 10 cm, height 4 cm

Method 1 (rebuild): split it into a rectangle 6 cm by 4 cm and a triangle with base 4 cm and height 4 cm.

$$6 \times 4 + \frac{1}{2} \times 4 \times 4 = 24 + 8 = 32 \text{ cm}^2$$

Method 2 (formula):

$$A = \frac{1}{2}(a + b)h = \frac{1}{2}(6 + 10) \times 4 = 32 \text{ cm}^2$$

Both methods agree, and learners see where the formula comes from.

EXAMPLE 6 Angles of a pentagon

From one corner of a pentagon, draw lines to the other corners. The pentagon is split into 3 triangles, and each triangle has angles adding up to 180°.

$3 \times 180^\circ = 540^\circ$ (the formula is $(n - 2) \times 180^\circ$)

EXAMPLE 7 Simple interest: GH¢200 at 5% per year for 3 years

Year by year: 5% of GH¢200 is GH¢10 each year, so for 3 years the interest is 3 × GH¢10 = GH¢30.

Check with the formula: $I = \frac{P \times R \times T}{100} = \frac{200 \times 5 \times 3}{100} = 30$

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3 Show more than one method

When a learner gets an answer, ask: “Who solved it a different way?” Put two or three methods side by side. Learners learn to choose the method that suits the numbers.

EXAMPLE 8 Find 15% of GH¢80

Method 1 (build up): 10% of 80 = 8 and 5% of 80 = 4, so 15% = 8 + 4 = 12.

Method 2 (decimal): 0.15 × 80 = 12.

Method 3 (fraction):

$$\frac{15}{100} \times 80 = \frac{3}{20} \times 80 = 12$$

Answer: GH¢12 by all three methods.

EXAMPLE 9 Calculate 25 × 12

Method 1: 25 × 10 + 25 × 2 = 250 + 50 = 300.

Method 2: 25 × 4 × 3 = 100 × 3 = 300.

EXAMPLE 10 Solve 3x + 5 = 20

Method 1 (balance): take 5 from both sides, 3x = 15, then divide both sides by 3, so x = 5.

Method 2 (work backwards): 20 − 5 = 15, then 15 ÷ 3 = 5.

Method 3 (trial and check): try x = 5: 3(5) + 5 = 20 ✓.

4 Move from concrete to picture to symbols

Let learners handle or draw the idea first. Only then write the rule. Weak learners often fail not because mathematics is hard, but because they were shown symbols too soon.

EXAMPLE 11 Dividing fractions: how many sixths are in two-thirds?

Concrete: fold a strip of paper into 3 equal parts, then fold each part into 2. Shade 2 of the 3 parts: that is 4 of the small sixths.

Symbols:

$$\frac{2}{3} \div \frac{1}{6} = \frac{2}{3} \times \frac{6}{1} = \frac{12}{3} = 4$$

EXAMPLE 12 Integers on a number line

Picture: the temperature is −3°C and it rises by 5°C. Start at −3 on a number line and move 5 steps to the right. You stop at 2.

$$(-3) + 5 = 2$$

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EXAMPLE 13 Solving an equation with a balance scale: 2x + 3 = 11

Picture: a balance with two bags (x each) and 3 loose marbles on one side, and 11 marbles on the other. Take 3 marbles from each side. Two bags balance 8 marbles, so one bag balances 4.

$$2x + 3 = 11 \Rightarrow 2x = 8 \Rightarrow x = 4$$

5 Estimate first, then check that the answer makes sense

A quick estimate protects learners from silly errors such as a misplaced decimal point. Make “Is my answer reasonable?” a habit before any answer is written in the box.

EXAMPLE 14 Multiplying 49 × 21

Estimate: 50 × 20 = 1,000. The exact answer is 1,029, which is close to the estimate, so it is reasonable.

EXAMPLE 15 Three exercise books at GH¢4.80 each

Estimate: 3 × GH¢5 = GH¢15. The exact cost is 3 × 4.80 = GH¢14.40. A learner who writes GH¢144 can see at once that the decimal point is wrong.

EXAMPLE 16 How big is √50?

$7^2 = 49$ and $8^2 = 64$, so $\sqrt{50}$ is a little more than 7. A calculator gives 7.07 correct to 2 decimal places.

EXAMPLE 17 Checking an addition of fractions

Ama writes $\frac{3}{5}+\frac{2}{5}=\frac{5}{10}$. Estimate: $\frac{3}{5}$ is already more than half, so adding more cannot give exactly half. The correct answer is $\frac{5}{5}=1$.

6 Use “spot the mistake” tasks

Show a worked solution that contains a typical error and ask learners to find it, explain it and correct it. Errors discussed openly stop being embarrassing and stop being repeated.

EXAMPLE 18 Kofi writes 2(x + 3) = 2x + 3

Test with x = 1. Left side: 2(1 + 3) = 8. Right side: 2(1) + 3 = 5. They are not equal.

$$\text{Correct: } 2(x+3)=2x+6$$

EXAMPLE 19 Ama writes (a + b)² = a² + b²

Test with a = 3 and b = 4. Left side: $(3 + 4)^2 = 49$. Right side: 9 + 16 = 25.

$$\text{Correct: } (a+b)^2=a^2+2ab+b^2$$

EXAMPLE 20 Esi says 0.25 is bigger than 0.5 because 25 is bigger than 5

Write 0.5 as 0.50, or think of money: GH¢0.50 (50 pesewas) is more than GH¢0.25 (25 pesewas). So 0.5 > 0.25.

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EXAMPLE 21 Yaw writes √9 + √16 = √25

$\sqrt{9}+\sqrt{16}=3+4=7$, but $\sqrt{25}=5$. A square root does not split over addition.

EXAMPLE 22 Brackets matter: −3² and (−3)²

$$-3^2=-(3\times 3)=-9 \qquad \text{but} \qquad (-3)^2=(-3)\times(-3)=9$$

7 Use real situations from the learners’ own world

Learners engage when problems come from the market, the lorry station, the mobile money shop and the farm. Collect real prices, distances and quantities from the community.

EXAMPLE 23 Profit percent at the market

A trader buys a bag of rice at GH¢350 and sells it at GH¢420. Profit = GH¢420 − GH¢350 = GH¢70.

$$\text{Profit }\% = \frac{70}{350}\times 100 = 20\%$$

EXAMPLE 24 Commission of a mobile money agent

An agent earns 1.5% commission on transactions worth GH¢2,000.

$$\frac{1.5}{100}\times 2000 = 30$$

The commission is GH¢30.

EXAMPLE 25 Ratio: mixing paint

White and blue paint are mixed in the ratio 3 : 2 to make 20 litres. Total parts = 5, so 1 part = 20 ÷ 5 = 4 litres.

White = 3 × 4 = 12 litres. Blue = 2 × 4 = 8 litres.

EXAMPLE 26 Rate: fuel used by a car

A car uses 8 litres of fuel for 100 km. How much fuel does it use for 36 km?

$$8 \times \frac{36}{100} = 2.88 \text{ litres}$$

EXAMPLE 27 Statistics with the class: shoe sizes of 10 learners

Sizes: 5, 6, 6, 7, 7, 7, 8, 8, 9, 9. The total is 72, so the **mean** = 72 ÷ 10 = 7.2. The **mode** (most common) is 7. The **median** (middle) is 7, because the 5th and 6th values are both 7.

8 Take mathematics outside the classroom

A tape measure, a ball of string, a metre rule and a sunny day can teach more mathematics than a week of board work. Real objects give learners something to see and test.

EXAMPLE 28 Tiling the classroom floor

The floor measures 8 m by 6 m, so its area is 8 × 6 = 48 m². Each tile is 0.5 m by 0.5 m, an area of 0.25 m².

Tiles needed = 48 ÷ 0.25 = 192

At GH¢6 per tile, the cost is 192 × GH¢6 = GH¢1,152.

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EXAMPLE 29 Height of the flagpole using shadows

At the same time of day, a 1 m stick casts a shadow of 1.5 m and the flagpole casts a shadow of 9 m. The two triangles have the same shape.

$$\frac{h}{9} = \frac{1}{1.5} \Rightarrow h = \frac{9}{1.5} = 6 \text{ m}$$

EXAMPLE 30 Right angles with a 3-4-5 rope (Pythagoras)

Carpenters mark a rope into 3, 4 and 5 equal units and make a triangle. The corner between the sides 3 and 4 is a right angle, because $3^2 + 4^2 = 5^2$.

Use it: a 5 m ladder stands with its foot 3 m from a wall. How high up the wall does it reach?

$$h = \sqrt{5^2 - 3^2} = \sqrt{25 - 9} = \sqrt{16} = 4 \text{ m}$$

9 Ask questions that make learners think and talk

The questions a teacher asks decide the kind of thinking that happens. Replace “Is that right?” with prompts that make learners explain.

PROMPTS "How do you know?" · "Can you show me another way?" · "What would happen if...?" · "Is it always, sometimes or never true?" · "What is the same and what is different?" · "Can you draw it?"

EXAMPLE 31 Always, sometimes or never true?

"A square is a rectangle." Always true, because a square has four right angles and opposite sides equal.

"Multiplying a number makes it bigger." Sometimes true: 8 × 2 = 16 is bigger, but 8 × 0.5 = 4 is smaller.

"Adding two even numbers gives an even number." Always true: 6 + 10 = 16, 14 + 2 = 16 and so on.

EXAMPLE 32 Odd one out: 16, 25, 36, 48

Learners may choose **48**, because it is not a perfect square. Another learner may choose **25**, because it is the only odd number. Both answers are correct when the reason is given, and this is exactly the kind of talk we want.

10 Give open-ended tasks with more than one answer

Open tasks let weak learners start and strong learners go deeper. They also show that mathematics is about reasoning, not guessing what is in the teacher's head.

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EXAMPLE 33 Rectangles with an area of 24 cm² (whole-number sides)

1 × 24 has perimeter 50 cm, 2 × 12 has 28 cm, 3 × 8 has 22 cm, and 4 × 6 has 20 cm. Same area, different perimeters, so area and perimeter are not the same thing.

EXAMPLE 34 Rectangles with a perimeter of 20 cm

1 × 9 has area 9 cm², 2 × 8 has 16 cm², 3 × 7 has 21 cm², 4 × 6 has 24 cm² and 5 × 5 has 25 cm². The square gives the largest area.

EXAMPLE 35 Make up your own question

"Write a question whose answer is 36." Learners write problems such as 6^2, 4 × 9, 100 − 64, or "the area of a square of side 6 m". Share the best ones with the class.

11 Start lessons with five minutes of mental mathematics

Short daily starters build number sense and confidence. Ask learners to explain how they got the answer, not only what it is.

EXAMPLE 36 Four quick starters

98 + 47: 100 + 47 − 2 = 145.

6 × 15: 6 × 10 + 6 × 5 = 60 + 30 = 90.

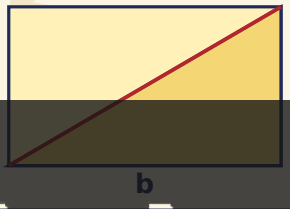
99 × 7: 100 × 7 − 7 = 700 − 7 = 693.

$7^2 - 6^2$: notice that $7^2 - 6^2 = (7 + 6)(7 - 6) = 13 \times 1 = 13$.

12 Let learners keep a formula notebook and teach each other

Each formula gets one page in the learner's own words: a picture, where it comes from, and a worked example. Learners who can teach a formula to a classmate understand it.

A sample page from a formula notebook:

Name	Area of a triangle
Picture	
In words	Half of the base times the perpendicular height.
Formula	$A = \frac{1}{2} b h$
Why it works	Two equal triangles fit together to make a rectangle of area b × h, so one triangle is half of it.
Worked example	b = 10 cm and h = 6 cm. $A = \frac{1}{2} \times 10 \times 6 = 30$ cm^2.
Check another way	The rectangle is 10 × 6 = 60 cm², and half of 60 is 30 cm² □.

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Common misconceptions and quick fixes

Learners carry some wrong ideas into JHS. Name them, test them with numbers, and fix them with a picture or a real situation.

Misconception	Quick fix	Check with
"Multiplying always makes a number bigger."	Multiply by a number less than 1 and the answer is smaller.	8 × 0.5 = 4
"0.25 is bigger than 0.5 because 25 > 5."	Compare equal place values: write 0.5 as 0.50.	0.50 > 0.25
"The bigger the denominator, the bigger the fraction."	More pieces means smaller pieces.	$\frac{1}{8} < \frac{1}{4}$

Misconception	Quick fix	Check with
"Adding fractions: add tops and add bottoms."	Use a common denominator first.	$\frac{1}{2}+\frac{1}{3}=\frac{3}{6}+\frac{2}{6}=\frac{5}{6}$
"−7 is greater than −2 because 7 > 2."	Use a number line or temperatures: −7°C is colder.	$-7 < -2$
"3 − 7 = 4, because we subtract the smaller from the larger."	Order matters. Show it on a number line.	$3 - 7 = -4$
"Perimeter and area are the same thing."	A fence measures perimeter; the grass inside measures area.	Area 24 cm² can have perimeter 20 cm or 50 cm
"$(a + b)^2 = a^2 + b^2$."	Test with numbers, then expand fully.	$(3 + 4)^2 = 49$, but $3^2 + 4^2 = 25$
"In 3x, the letter x is a label, like '3 mangoes'."	x stands for a number, and 3x means 3 × x.	If x = 4, then 3x = 12
"Zeros at the front of 0.0046 count as significant figures."	Count from the first non-zero digit.	0.0046 has 2 significant figures

A rescue plan for learners who forget a formula

Teach learners a calm routine to follow when memory fails in a test or examination:

1. **Stay calm and read again.** Ask: what is the question really asking for?
2. **Draw it.** A quick sketch often shows what to do.
3. **Try small numbers.** Solve an easier version first and look for the pattern.
4. **Use another method.** Count squares, split the shape, work year by year, or work backwards.
5. **Check.** Does the answer make sense? Are the units right?

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A sample 40-minute lesson: the area of a triangle

Time	Stage	What happens
0–5 min	Starter	Recall: find the area of a rectangle 8 cm by 5 cm (40 cm²). Ask how they know.
5–15 min	Discover	Pairs cut a rectangle along a diagonal and lay one piece on top of the other. Ask: what fraction of the rectangle is each triangle?
15–22 min	Build the rule	Label base b and height h. The rectangle has area b × h, so each triangle has $A = \frac{1}{2}bh$. Learners write it in their formula notebooks.
22–32 min	Practise	(1) b = 12 cm, h = 5 cm: $\frac{1}{2} \times 12 \times 5 = 30$ cm². (2) A triangular garden has base 14 m and height 9 m: $\frac{1}{2} \times 14 \times 9 = 63$ m². (3) Area 36 cm² and base 9 cm. Find the height: $\frac{1}{2} \times 9 \times h = 36$, so $4.5h = 36$ and h = 8 cm.
32–38 min	Think	Draw three different triangles with an area of 12 cm². For example, b = 6 and h = 4, or b = 8 and h = 3, or b = 12 and h = 2.

Time	Stage	What happens
38–40 min	Exit ticket	In one sentence, explain why the area of a triangle is half of base times height.

A weekly checklist for the mathematics teacher

- ☐ Start each lesson with a five-minute mental mathematics or recall starter.
- ☐ Introduce at least one new formula this week by discovery, not by telling.
- ☐ Give one “spot the mistake” task and discuss it openly.
- ☐ Ask “How do you know?” at least five times in every lesson.
- ☐ Use one example from the learners’ own community (market, farm, transport, mobile money).
- ☐ End a lesson with an exit question that asks learners to explain, not just calculate.
- ☐ Let learners add one page to their formula notebook.

The lesson every learner should remember

Imagine two learners sitting for the same mathematics examination.

THE FIRST LEARNER

The first learner has memorised many formulas but does not understand them. When a question is presented in an unfamiliar way, the learner struggles to identify the correct formula.

THE SECOND LEARNER

The second learner understands the concepts, recognises the relationships involved and knows how to reason through a problem. Even when a formula is forgotten, the learner can explore another approach.

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Who is better prepared for mathematical problem-solving? **The second learner**, because understanding provides a foundation for flexible thinking.

The ultimate goal of mathematics education is not to produce learners who can only recall formulas. It is to develop learners who can think mathematically, explain their reasoning and solve unfamiliar problems.

Memorise formulas, yes—but never allow memorisation to replace understanding.

Mathematics is not just about knowing what to do. It is about understanding why it works.

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THE WRITER

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